

PABLO VILLANUEVA DOMINGO

PHD IN PHYSICS & DEEP LEARNING SCIENTIST

AI scientist specializing in **deep learning** and **multi-agent systems**, with a **PhD in theoretical physics** (*cum laude*, University of València). Currently developing RL pipelines and autonomous agents for the CARLA simulator (UAB/CVC) and scientific discovery platforms. Background in applying deep learning to cosmology and astrophysics; 23 publications, h-index 17, 1300+ citations.



CONTACT

✉ Pablo.Villanueva.Domingo@gmail.com
🏠 pablovd.github.io
🐙 @PabloVD
🌐 pablo-villanueva-domingo-76b6491b2
📞 0000-0002-0936-4279
📄 [Pablo Villanueva Domingo](#)
🐦 @CosmoPabloVD

SKILLS

💻 Computation

Programming languages

Python, Typescript, C++, C, C#, Fortran, SQL

Web/App development

React, React Native, Tailwind, Jekyll, FastAPI

General software

Git, Docker, LaTeX, MATLAB, Mathematica

Data analysis

Numpy, SciPy, Pandas, OpenCV, Networkx

Visualization

Matplotlib, Seaborn, Plotly, Folium

Simulation & game engines

Unreal, Godot, Chrono, Flex, Unity

Autonomous driving & robotics software

CARLA, IsaacLab, ROS, Scenic, Xviz

🤖 Machine learning

ML libraries

PyTorch, TensorFlow, PyTorch Lightning,

PyTorch Geometric, Scikit-learn

Neural Nets Architectures

Graph (GNNs), Convolutional (CNNs), U-Nets,

Diffusion models, GANs, LSTMs

Agents & LLMs

Langchain/Langgraph, AG2, Denario, RAGs

Fields

Computer vision, Natural Language

Processing, Reinforcement Learning

See my work in ML and programming at

<https://pablovd.github.io/codes>

💬 Soft skills

Communication

Public speaking, writing skills

Project management

Collaboration, teamwork, initiative, organization

Problem solving

Logical reasoning, lateral thinking, creativity, data modeling

🌐 Languages

Spanish	Mother tongue
Catalan	Mother tongue
English	Fluent
Portuguese	Basics

📁 WORK HISTORY

• Deep Learning Scientist

📅 Jan. 2022- Now | 📍 Computer Vision Center - Universitat Autònoma de Barcelona
Reinforcement learning training, Data-driven traffic models, neural terramechanics and computer vision tasks at the autonomous driving simulator [CARLA](#)

• Research assistant

📅 Jun. 2021- Dec. 2021 | 📍 Instituto de Física Corpuscular - Universitat de València
Técnico superior de apoyo a la investigación, CIDEAGENT/2018/019, CPI-21-108

• PhD fellowship

📅 Oct 2016 - Mar. 2021 | 📍 Instituto de Física Corpuscular - Universitat de València
FPI Severo Ochoa, Ref. SEV-2014-0398-16-3

• Research introduction fellowship

📅 May-Oct. 2016 | 📍 Instituto de Física Corpuscular
Iniciación a la investigación Severo Ochoa

🔗 MAJOR PROJECTS

• [Denario](#) – Multi-agent AI Scientist Platform

📅 2024 – Now | 📍 astropilot-ai.github.io/DenarioPaperPage
Co-creator of a multi-agent system for AI assistance of research and end-to-end scientific discovery, spanning 12 disciplines. 36 co-authors and 500+ GitHub stars.

• [CARLA](#) – Autonomous Driving Simulator

📅 Jan. 2022 – Now | 📍 carla.org
Development of machine learning pipelines, data-driven traffic models, and computer vision modules for the open-source CARLA simulator.

• [AeroSim](#) – Aviation Simulation Platform

📅 2024 – Now | 📍 github.com/aerosim-open/aerosim
Development of the open-source AeroSim simulation platform and its visual user interface for aerospace applications.

🎓 EDUCATION

• PhD in Physics, *cum laude*

📅 2016-2021 | 📍 Instituto de Física Corpuscular - Universitat de València

• Master in Advanced Physics

📅 2015-2016 | 📍 Universitat de València

• Bachelor of Physics

📅 2011-2015 | 📍 Universitat de València

As well as multiple PhD schools and courses which can be found [here](#).

🏆 AWARDS

📅 Feb. 2023 | CSIC 2021 relevant PhD Thesis Award, by Consejo Superior de Investigaciones Científicas (CSIC).

📅 Dec. 2016 | 1st prize in the XXVII edición del Premio Rotary al Fomento del Trabajo Experimental en Física.

✈ RESEARCH STAYS

I have participated in several international research collaborations, visiting universities from different countries:

- 📅 Nov.- Dec. 2019 | 📍 3 weeks at Service de Physique Théorique, Université Libre de Bruxelles, Brussels, Belgium.
- 📅 Sep.- Oct. 2019 | 📍 1 month at Department of Astrophysical Sciences, Princeton University, New Jersey, USA.
- 📅 Sep.- Nov. 2018 | 📍 2 months at Kavli IPMU, University of Tokyo, Japan.
- 📅 Jun.- Aug. 2017 | 📍 2 months at Fermi National Accelerator Laboratory (Fermilab), Illinois, USA.

🗣 TALKS

I have delivered **9 invited seminars** at the universities of Princeton (USA), Tokyo, Nagoya (Japan), Brussels and València, and presented **8 talks** at international conferences, meetings and schools.

A complete list can be found at <https://pablovd.github.io/talks.pdf>. These are some selected talks:

- *Neural Terramechanics and the RACER-SIM project*

📅 Jul. 28 2022 | 📍 EAI Tech Forum, Intel Labs, online

- *Weighing the Milky Way with AI*

📅 Jan. 17 2022 | 📍 *Cosmology Talks*, online (Youtube channel) | [Video](#)

- *Machine Learning at galactic and cosmological scales*

📅 Nov. 17 2021 | 📍 Instituto de Física Corpuscular | [Video](#) and [slides](#)

📄 SELECTED PUBLICATIONS

I have published **23 scientific articles** in high impact journals based on my research on cosmology, deep learning and multi agent systems, with over **1300 citations** and an **h-index of 17** ([Google Scholar](#)). The full list of publications can be found in my INSPIRE profile [P.Villanueva.Domingo.1](#). These are some selected publications:

- *The Denario project: Deep knowledge AI agents for scientific discovery*

Francisco Villaescusa-Navarro, Boris Bolliet, **Pablo Villanueva-Domingo**, Adrian E. Bayer, et al.

📅 Oct. 2025 | 📄 [arXiv:2510.26887](#)

A multi-agent AI system for assistance in scientific research, spanning multiple fields from physics to medicine.

- *Weighing the Milky Way and Andromeda with Artificial Intelligence*

Pablo Villanueva-Domingo, Francisco Villaescusa-Navarro, Shy Genel, Daniel Anglés-Alcázar, Lars Hernquist, Federico Marinacci, David N. Spergel, Mark Vogelsberger and Desika Narayanan

📅 Nov. 2021 | 📄 [Physical Review D 107, 103003, 2023](#), [arXiv:2111.14874](#)

The total masses of the Milky Way and Andromeda galaxies are predicted using AI for the first time, via Graph Neural Networks.

- *Removing Astrophysics in 21 cm maps with Neural Networks*

Pablo Villanueva-Domingo and Francisco Villaescusa-Navarro

📅 Jan. 2021 | 📄 [The Astrophysical Journal, 907\(1\):44, 2021](#); [arXiv:2006.14305](#)

The cosmic density field is predicted from maps of distribution of hydrogen training a U-Net in PyTorch.

⚙ OUTREACH & ADDITIONAL WORK EXPERIENCE

📅 Feb. 2021 | [Outreach video](#) about the astronomer Sandra M. Faber within the project *Pioneras - Recordando a Lise Meitner*.

📅 2020 - Now | Journal referee for journals such as MNRAS and ApJ. See reviews in my [Publons](#) profile

📅 Jun. 2019 | Member of the local organizing committee of the Invisibles19 Workshop at València and Invisibles19 School at Laboratorio subterráneo de Canfranc (LSC)

📅 2016-2017 | Collaboration in the organization of the outreach event *Feria-Concurso Experimental*, València.